Logic over Wiring: Sign-Aware Graph Neural Networks for Asynchronous Activation Reachability in Boolean Gene-Regulatory Networks

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***Abstract*—Boolean networks (BNs) are a standard qualitative model of gene-regulatory and signalling dynamics, in which the behaviour of interest is captured by attractors and by reachability between states. Both are hard to compute exactly in general. Classical theory connects a BN's interaction graph to its dynamics only through necessary conditions and bounds, so the wiring diagram provably under-determines the attractor set. We use this gap to pose a well-defined learning problem with free, exact supervision. With an exact symbolic oracle supplying ground-truth labels, we first measure the gap itself: a predictor restricted to graph topology performs at chance (AUROC 0.56), because fixing the wiring and varying only the Boolean functions flips the attractor label for 84–88% of wirings. We then introduce a sign-aware, size-agnostic graph neural network (GNN) that reads the local Boolean logic and predicts *per-node asynchronous activation reachability*, that is, which genes can switch on from a quiescent state. In distribution, the GNN beats a gradient-boosted tree given *all* truth-table bits on reachability (0.96 vs 0.92), but not on the more global has-cyclic property; we attribute this selective win to an inductive bias rather than to capacity. Trained only on small synthetic BNs, the same GNN then transfers *zero-shot* to real curated gene-regulatory networks from the biodivine-boolean-models collection, reaching pooled AUROC 0.91 (95% CI $[0.88, 0.93]$) and outperforming every size-agnostic baseline, including a CANA effective-connectivity baseline whose local-feature mapping fails to transfer. A sign / no-sign / MLP ablation traces the gain to sign-aware message passing rather than to recomputed canalization. Because all labels come from exact symbolic computation, the method needs no manual annotation and runs on a single GPU.**

***Index Terms*—Boolean networks, gene regulatory networks, graph neural networks, attractors, asynchronous reachability, canalization, machine learning, systems biology.**

I. Introduction

How much of a gene-regulatory network's long-run behaviour is fixed by its *wiring*, and how much by the regulatory *logic* on that wiring? Boolean networks (BNs) are a canonical qualitative model of gene regulation, signalling, and cell-fate decision-making: each gene is a binary node, and a Boolean update function of its regulators encodes the combinatorial logic of activation and repression [1], [2]. The biologically meaningful predictions of a BN are its *attractors*, the long-run cyclic or steady patterns identified with cell phenotypes, together with its *reachability* structure: from a given state, which configurations and which individual gene activations the dynamics can attain. A question that recurs across systems biology is which genes can be switched on from a quiescent (all-OFF) state, and by what regulatory routes. It underlies reprogramming and signal-response analysis.

On the structural side the theory is sharp. Thomas's rules tie multistationarity to positive feedback circuits and sustained oscillation to negative circuits [3], [4]; Aracena bounds the number of fixed points by the positive feedback vertex set [5]. Yet these are *necessary conditions and bounds* rather than determinations. The same signed interaction graph admits update functions with one attractor or with many, and deciding the count from the graph alone is #P-complete [6]. The interaction graph, in short, does not fix the dynamics; the logic does.

This under-determination is what turns prediction into a genuine *learning* problem (Fig. 1). If the graph does not determine the dynamics, then a model reading the graph alone can do no better than chance, and any model that succeeds must be drawing on the *logic*. Predicting a dynamical property from structure and logic together is therefore a well-posed supervised task, with a built-in identifiability guarantee and a free, exact label source in the symbolic oracle, and with no need for manual annotation. We instantiate it for *asynchronous reachability*. This property is PSPACE-hard to compute exactly in general, it is biologically central, and, unlike the mere existence of attractors, it is fundamentally a *propagation* phenomenon.

Our learner is a sign-aware, size-agnostic GNN that reads each node's local Boolean function (through arity-invariant descriptors) and the signed wiring, and predicts per-node activation reachability. Being size-agnostic, it is trained once on cheap small synthetic networks and applied to real gene-regulatory networks (GRNs) it never saw.

Contributions.

- *An identifiability control.* An exact-oracle, de-confounded quantification that the interaction graph under-determines BN attractor properties on our distribution (84–88% label flips under fixed wiring; topology-only AUROC 0.56), certifying that a learned signal must come from the logic (§V).
- *A logic-aware, sign-aware, size-agnostic GNN* predicting asynchronous reachability that, in distribution, beats a gradient-boosted tree given *all* truth-table bits specifically on reachability and not on the global has-cyclic property;

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this reflects an inductive bias rather than raw capacity. A sign / no-sign / MLP ablation and a dedicated CANA effective-connectivity baseline isolate sign-aware propagation as the source of the gain (§VI, §VII-C).

- *Zero-shot transfer to real GRNs.* Trained only on synthetic BNs, the GNN predicts per-node activation reachability on real biodivine-boolean-models networks at pooled AUROC 0.91 (95% CI [0.88, 0.93]), beating size-agnostic feature baselines by $+0.09$ and a CANA baseline that collapses below chance on transfer (§VII-B).
- *A reproducible, annotation-free pipeline*: an exact state-transition-graph oracle for synthetic networks and symbolic BDD reachability for real ones, on a single GPU.

II. Background

Boolean networks. A BN on n nodes assigns each node i a Boolean update function $f_i : \{0,1\}^{k_i} \to \{0,1\}$ of its k_i *regulators*. A global state is $x \in \{0,1\}^n$. The *signed interaction graph* has an edge $j \to i$ whenever f_i depends on x_j; the edge is labelled $+$ $(-)$ if f_i is locally monotone increasing (decreasing) in x_j, and unsigned if non-monotone. The graph thus records *who regulates whom and with what sign*, but discards the combinatorial logic of how regulators combine.

Updating schemes and attractors. Under *synchronous* update all nodes update simultaneously, $x \mapsto F(x) = (f_1(x), \ldots, f_n(x))$; the dynamics is a deterministic functional graph whose cycles are the synchronous *attractors* (fixed points have period 1). Under *asynchronous* update one node updates at a time, yielding a non-deterministic state-transition graph (STG) whose terminal strongly connected components are the asynchronous attractors. Asynchronous update is the more biologically faithful regime, avoiding artificial synchrony, but its STG has out-degree up to n and reachability questions are correspondingly harder (PSPACE-hard in general).

Reachability. The *forward-reachable set* of a state x under asynchronous dynamics is the set of states attainable by any sequence of single-node updates. We focus on *per-node activation reachability*: starting from the all-OFF state $\mathbf{0}$, can node v ever take value 1? Biologically this asks which genes are activatable from quiescence; it is a node-classification target and is governed by whether activation can *propagate* along the signed wiring.

Canalization. A Boolean function is canalizing when some input value alone fixes the output; nested canalization and the associated *effective connectivity* k_e and *input redundancy* measure how few inputs effectively matter [12]. Real GRN functions are strongly canalizing, which shapes their dynamics; we therefore treat canalization features as a strong baseline to beat (§VII-C).

Why this is a learning problem. Counting fixed points is #P-complete and the interaction graph only bounds it [5], [6]; the graph under-determines attractors and reachability (§V). Hence predicting these properties from structure-plus-logic is non-trivial precisely where the graph is uninformative. We emphasize that our aim is *not* to replace the exact symbolic solver on a speed basis (on the curated networks studied here the symbolic oracle is itself fast), but to establish that the logic carries *transferable* reachability structure: a single learned model generalizes across networks it never saw, whereas an exact solver must recompute each network from scratch and learns nothing portable.

III. Related Work

Structure→dynamics theory. Thomas's conjectures are proved by Richard and Comet [3] and are necessary, not sufficient; the same signed graph supports very different dynamics depending only on the logic [4]. Aracena bounds the fixed-point count by 2^{τ^+} [5]; deciding k-fixed-points from the digraph is #P-complete [6], and the regulatory logic, not the wiring, shapes the state-transition graph [7]. We quantify this gap as a control; we do not claim it as new.

Predicting BN dynamics. Rumiantsau *et al.* [8] predict BN attractors from structure via spectral analysis, which is non-learned and targets fixed-point patterns, with no asynchronous reachability and no transfer. Weidner *et al.* [9] rank node dynamic-impact from graph-theoretic features (a forest on graph measures); this is exactly our topology baseline, which we show is near-chance for reachability. GATTACA [10] pairs a GNN with reinforcement learning to *control* cellular reprogramming—learning which genes to perturb to steer the network toward a target attractor—a control task rather than forward-dynamics prediction, and one that does not predict per-node reachability. To our knowledge, no prior work learns to predict asynchronous reachability of BNs from local logic, and none demonstrates synthetic→real transfer.

Learning discrete dynamics; canalization. Methods that learn *unknown* local rules from observed trajectories [11] address the inverse, system-identification problem and do not read a known Boolean logic; ML for "Boolean control network reachability" is uniformly reinforcement-learning *control*; and ML for GRNs is overwhelmingly network *inference* (the opposite direction to forward dynamics prediction). The strongest non-trivial baseline is canalization: effective connectivity and input redundancy (CANA) [12] are known local logic features that strongly shape BN dynamics; §VII-C shows our gain is not recomputed canalization.

Data and oracle. We use the biodivine-boolean-models (BBM) collection of curated biological BNs, with exact attractor/reachability analysis from `biodivine_aeon` and `biobalm` (symbolic, BDD-based) [13], [14], [15], repurposed here as a zero-cost supervision source.

IV. Tasks and the Exact Oracle

Prediction tasks. We study three exact-oracle labels: (a) *has-cyclic-attractor*, whether a synchronous attractor of period > 1 exists; (b) *attractor count*; and (c), our headline task, *per-node activation reachability* as defined in §II. Tasks (a)–(b) are global network-level properties; task (c) is node-level and propagation-structured.

Synthetic oracle. For small networks ($n \leq 16$) we compute labels exactly by enumerating the STG. Synchronous attractors are the cycles of the functional graph (Alg. 1), found by a

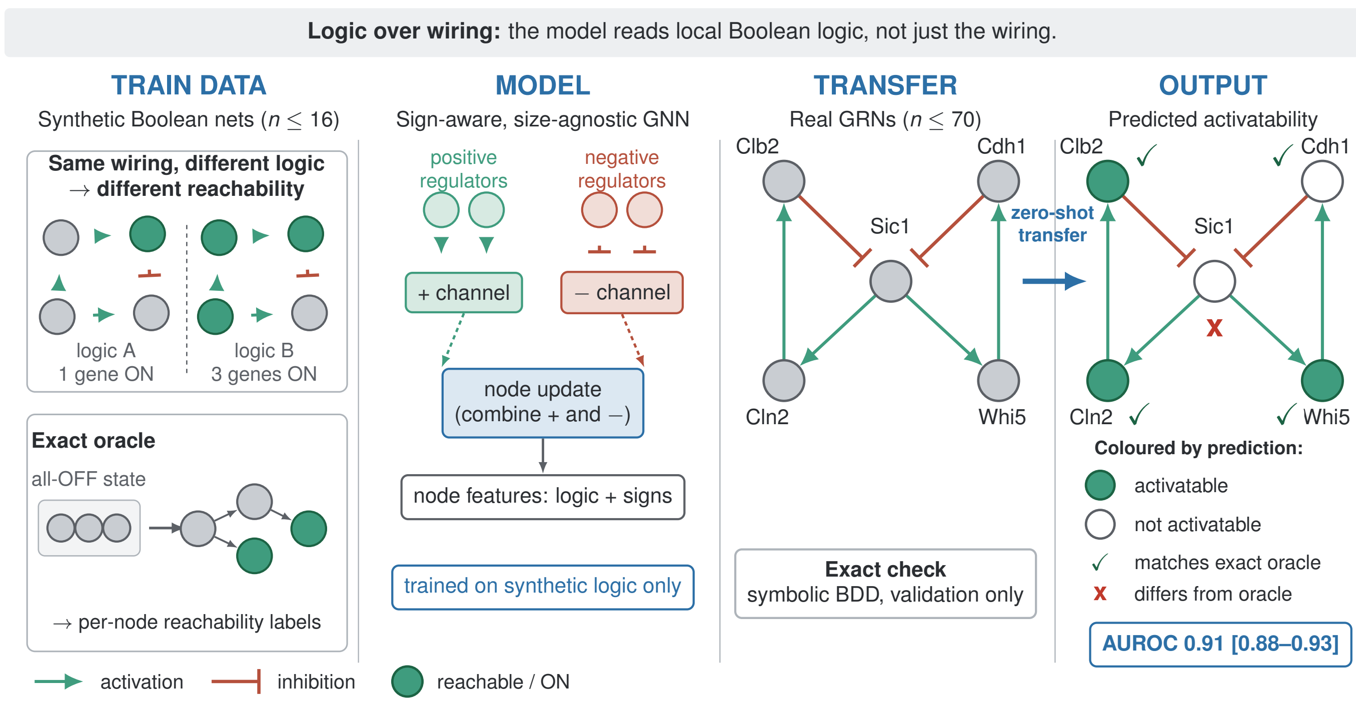

Fig. 1. **Overview.** *Train data*: an exact state-transition-graph oracle labels small synthetic Boolean networks ($n \leq 16$) with per-node asynchronous activation reachability from the all-OFF quiescent state. The key inset shows why this is non-trivial: fixing the wiring and changing only the Boolean logic changes the reachable set. *Model*: a sign-aware, size-agnostic GNN reads arity-invariant logic features and edge signs, passing messages through separate transforms for activating (+) and inhibiting (−) regulators; it is trained on synthetic logic only, with no real-network labels. *Transfer*: the trained model is applied zero-shot to real gene-regulatory networks from the biodivine-boolean-models collection ($n \leq 70$), whose ground truth is obtained independently by symbolic BDD reachability. *Output*: predicted per-gene activatability from quiescence, validated against the exact oracle (pooled AUROC 0.91, 95% CI $[0.88, 0.93]$). Green arrows denote activation, red bars inhibition; filled green nodes are predicted activatable. Gene names are illustrative.

Algorithm 1 Exact synchronous attractors

$\text{succ}[s] \leftarrow F(s)$ for all $s \in \{0,1\}^n$; col $\leftarrow 0$
for $s_0 \in \{0,1\}^n$ with $\text{col}[s_0] = 0$ **do**
  follow successors marking col $= 1$ until a marked state s is hit
  if s is on the current path **then** record the cycle $s \to \ldots \to s$
  end if
  mark the whole path col $= 2$
end for
return list of cycles (attractors); period $1 \Rightarrow$ fixed point

Algorithm 2 Per-node activation reachability (async, from $\mathbf{0}$)

seen $\leftarrow \{\mathbf{0}\}$; $Q \leftarrow [\mathbf{0}]$; $r \leftarrow \mathbf{0}$
while Q non-empty **do**
  $s \leftarrow Q.\text{pop}()$
  for $i = 1 \ldots n$ with $f_i(s) \neq s_i$ **do**
    $s' \leftarrow s$ with bit i flipped
    if $s' \notin$ seen **then** add s' to seen, Q; $r \leftarrow r \,|\, s'$
    end if
  end for
end while
return $\text{label}[v] = r_v$ for each node v

single colour-marked traversal in $O(2^n)$ time. Per-node activation reachability is computed by an asynchronous breadth-first search from $\mathbf{0}$, OR-accumulating all reachable states (Alg. 2); node v is labelled 1 iff bit v is set in the accumulated reachable set. Both are exact and deterministic.

Real-network oracle. For real networks explicit STG enumeration is infeasible, so we use `biodivine_aeon`'s symbolic, binary-decision-diagram (BDD) forward reachability, which represents reachable sets compactly and scales beyond 2^n for structured networks [13]. Real BBM models contain *external-input* nodes (no regulators, a free signal); we fix these to OFF, consistent with the all-OFF quiescent initial condition, so that reachability is deterministic and per-node labels are well defined. We then compute, per gene v, whether $v = 1$ is in the forward-reachable set from quiescence.

Datasets and coverage. Synthetic training data are exact-labelled small BNs ($n \in \{12, 14\}$, in-degree $K \in \{2, 3, 4\}$, random and nested-canalizing logic). For real evaluation we draw from the 285 curated BBM networks; after fixing inputs and keeping fully-specified networks with $n \leq 70$, 178 are labelled successfully (symbolic labelling is process-isolated with a hard timeout; 4 time out, 4 are parameterized), of which 118 have class variance and are used for AUROC (Table I).

V. Identifiability: The Graph Under-Determines the Dynamics

Graph under-determination is established theory (§II); we present a quantitative, de-confounded version solely as an

TABLE I
REAL-NETWORK (BBM) COVERAGE FOR EVALUATION.

quantity	value
curated BBM networks	285
fully-specified, $n \le 70$, labelled	178
with class variance (used for AUROC)	118
labelled nodes (class-varying nets)	3659
activation base rate (pooled)	0.34
skipped: parameterized / timeout	4 / 4

TABLE II
WITHIN-GRAPH AUROC BY FEATURE GROUP (DE-CONFOUNDED).

feature group	has-cyclic	async reachability
topology (wiring only)	0.558	0.558
signed graph (Thomas)	0.860	0.566
logic-aware (all truth-table bits)	0.803	0.921
shuffle-label null	0.520	0.515

identifiability control that makes the learning problem well-posed, certifying that any signal our GNN extracts comes from the logic rather than the wiring.

Design. We generate 480 fixed-wiring families ($n \in \{12, 14\}$, $K \in \{2, 3, 4\}$) and draw 8 logic variants (random and nested-canalizing) per family on the *same* wiring, labelling each exactly. A *within-graph* split places different logic variants of the same wirings in train and test. One detail matters here: we *balance* the random/canalizing mix across the split, because a naive variant-parity split inadvertently trains on all-random and tests on all-canalizing logic, leaking a distribution shift that artificially depresses scores. Balancing removes this confound.

Results. Holding the wiring fixed and varying only the logic flips the label for 84% of wirings (has-cyclic) and 88% (reachability). On these instances, any topology-only predictor is, by construction, near-chance. Empirically a topology predictor reaches AUROC 0.558, a shuffle-label null 0.52 (Table II, Fig. 2a). The signed graph (Thomas level) helps for has-cyclic (0.86) but remains at chance for reachability (0.566); only features that read the logic predict reachability (0.92). Thus the logic, not the wiring, carries the reachability signal.

Not merely chaos. One might worry the learnable residual is just the Kauffman–Derrida order–chaos transition. Table III controls for this: the structural/logic features beat the mean-field sensitivity λ (the Derrida slope [16]) by $+0.07$ AUROC, and in the ordered regime ($K = 1$) structure predicts the attractor count ($R^2 = 0.73$) where λ alone is uninformative ($R^2 = -0.05$). The learnable signal is genuine logic-dependent structure, not the chaos order parameter.

VI. METHOD

Arity-invariant node features. Each node is described by features defined for *any* in-degree, so the same encoder applies to synthetic nodes of in-degree 4 and real nodes of in-degree 15 (Table IV). None is a fixed-width truth table; for high-arity real nodes the function-derived features (bias, sensitivity, canalizing depth) are estimated by sampling, keeping per-node cost bounded. These features are thus computed exactly on the low-arity synthetic training nodes but sampled on high-arity transfer nodes, a mild train-to-test feature-estimation shift we note as a limitation.

TABLE III
CHAOS CONTROL: STRUCTURE VS. THE DERRIDA SENSITIVITY λ.

predictor	has-cyclic AUROC	count R^2 (K=1)
λ alone (Derrida)	0.799	-0.045
structural / logic features	0.872	0.727
gain beyond λ	$+0.073$	$+0.772$

TABLE IV
ARITY-INVARIANT PER-NODE FEATURES.

group	features
degree	in-degree, out-degree
function bias	mean output $\Pr[f{=}1]$
canalization	canalizing depth (# canalizing inputs)
sensitivity	mean and max single-input activity
edge signs	fractions activating / inhibiting / non-monotone

Sign-aware message passing. We separate incoming edges by sign and learn distinct transforms for positive and negative regulators. With A^+, A^- the positive/negative signed adjacency (mean-normalized, $\widehat{\cdot}$) and $H^{(\ell)}$ the layer-ℓ node states,

$$H^{(\ell+1)} = \sigma\big(W_s H^{(\ell)} + W_p\,\widehat{A^+} H^{(\ell)} + W_n\,\widehat{A^-} H^{(\ell)}\big). \quad (1)$$

Eq. (1) is a standard relational/signed message-passing layer in the style of R-GCN [17] and signed graph convolution [18], with the two relations being positive and negative regulation; we do not claim the architecture itself as novel. Our contribution is the *task framing* (predicting logic-dependent reachability where the graph is provably uninformative) and the *zero-shot synthetic→real transfer*, for which a sign-aware encoder is the natural choice and, as the ablation shows, a decisive one. We use three layers (64 hidden units) and a per-node sigmoid head (Alg. 3), implemented as sparse edge-list message passing (scatter-add) so it runs on networks of any size (trained at $n \le 16$, applied to n up to hundreds). Separating sign is biologically motivated: whether activation propagates to a node depends on whether its regulators activate or inhibit it, information that a sign-agnostic aggregator discards. The ablation in §VII-C confirms that this term is doing the work.

Baselines. We compare against three baselines, in increasing strength. *Flat-TT*: a gradient-boosted tree on the concatenation of *all* node truth-table bits plus the signed wiring, an upper bound on the logic information available, but applicable only at a fixed small size. *Feature tree*: a gradient-boosted tree on the same size-agnostic arity-invariant features the GNN uses (without the graph structure). *CANA tree*: a gradient-boosted tree on a full CANA-style canalization suite: per-input activity, effective connectivity k_e, input redundancy, activity distribution, canalizing depth, bias (§VII-C). The last directly tests whether the GNN merely recomputes canalization.

Training. The GNN is trained with binary cross-entropy (Adam, lr 2×10^{-3}, weight decay 10^{-5}, 60 epochs) on the synthetic node set; no real-network labels are used in training. All results are mean±std over 3 training seeds.

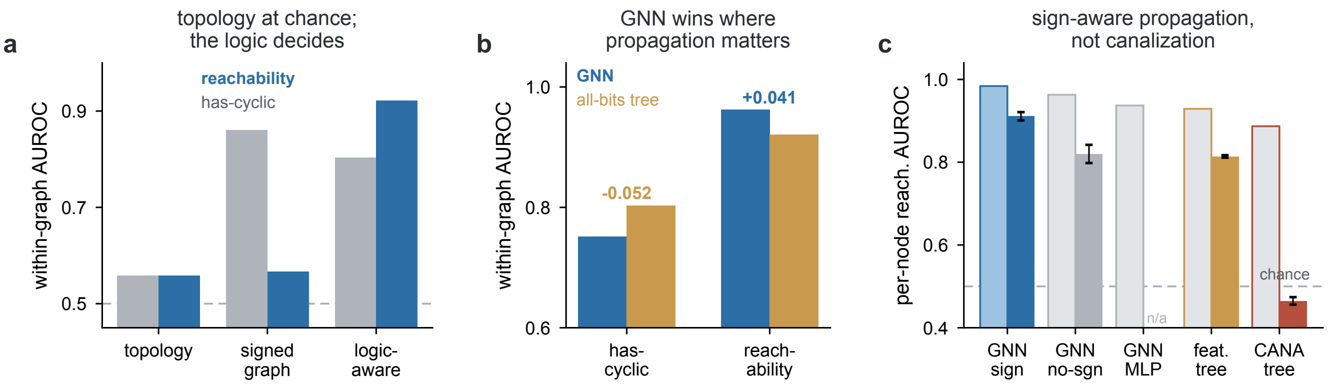

Fig. 2. The logic, not the wiring, carries learnable reachability structure. **a**, Within-graph AUROC by feature group (de-confounded): topology is at chance for both tasks, the signed graph helps only has-cyclic, and only logic-aware features predict reachability. **b**, The GNN's advantage is task-specific: it beats the all-bits tree on asynchronous reachability (propagation-shaped) but loses on the global has-cyclic property. **c**, Sign-aware propagation, not canalization: in-distribution (open bars) and zero-shot transfer (filled bars) AUROC for the GNN variants and the canalization baselines; the GNN-sign model (blue) stays high on transfer while the CANA tree drops below chance. Error bars: std over 3 seeds.

Algorithm 3 Sign-aware GNN forward pass (per-node head)

$H \leftarrow \mathrm{ReLU}(W_{\text{in}}X)$ ▷ X: arity-invariant features
for $\ell = 1 \dots L$ **do**
  $M^+ \leftarrow \widehat{A^+}H,\ M^- \leftarrow \widehat{A^-}H$ ▷ sign-split mean aggregation
  $H \leftarrow \mathrm{ReLU}(W_sH + W_pM^+ + W_nM^-)$
end for
return $\sigma(W_{\text{out}}H)$ ▷ per-node activation probability

VII. Experiments

Setup. Synthetic training and the 118-network real test set are as in §IV (Table I). We evaluate AUROC for the binary tasks.

A. In-distribution and the inductive-bias result

On the de-confounded within-graph split the flat-TT tree, which has *all* the logic bits, reaches 0.803 (has-cyclic) and 0.921 (reachability), far above the 0.52 null and generalizing leave-graph-out (0.819/0.916). The GNN beats the flat-TT tree on *reachability* (0.962 vs 0.921) but *loses* on the global *has-cyclic* property (0.751 vs 0.803, Fig. 2b); the sign-aware ablation harness of Table V, which uses a larger synthetic-training budget, reports higher absolute in-distribution values (0.984 vs 0.929) with the same ranking. A win on one task and a loss on the other points to an inductive bias rather than to greater capacity: message passing captures the propagation structure of reachability, whereas a feature-interaction tree does as well or better on a property that is not propagation-shaped. (This GNN-vs-tree comparison differs in input representation as well as architecture; the sign / no-sign / MLP ablation of §VII-C, which holds the input features fixed, isolates the architectural contribution.) The GNN is therefore suited specifically to reachability, and its cross-network generalization is established directly by the zero-shot transfer of §VII-B (a leave-graph-out test by construction).

B. Zero-shot transfer to real GRNs

We train the per-node reachability predictor on synthetic networks only, convert each real BBM network to the same representation, and evaluate zero-shot. The headline metric is the pooled AUROC over all 3659 nodes of the 118 real GRNs, with a *cluster* bootstrap that resamples whole networks (2000 replicates) to respect within-network correlation. The size-agnostic GNN attains pooled AUROC **0.906**, 95% CI $[0.879, 0.932]$, versus 0.816 $[0.794, 0.836]$ for the size-agnostic feature tree; the paired difference is $+0.091$, 95% CI $[0.064, 0.116]$, which excludes zero. The per-net AUROC distribution is left-skewed (median 0.99, IQR $[0.85, 1.00]$, mean 0.90; Fig. 3a); we report the pooled mean as the headline and the median only as a descriptor of the typical network. The flat-TT tree is *inapplicable* at these sizes (variable n and arity), and exact solvers do not generalize; they recompute per network.

Failure analysis. 12 of 118 networks fall below AUROC 0.7. Size alone does not cleanly predict this tail: across all 118 networks per-net AUROC shows no *significant* monotone correlation with size (Spearman $\rho = -0.14$, $p = 0.14$, 95% CI $[-0.31, +0.04]$; Fig. 3a, left), as most networks score near 1.0 regardless of size. The failures do, however, fall in the size-*extrapolation* regime: 10/12 have $n \geq 15$ (median tail size 21, worst case $n = 28$ at AUROC 0.222), i.e. larger than every synthetic training network ($n \leq 16$). The tail thus comprises larger networks whose wiring or sign structure is also atypical relative to the synthetic training distribution; characterizing these hard cases, and disentangling size extrapolation from structural atypicality, is left to future work.

C. The gain is sign-aware propagation, not canalization

One might object that reachability is already recoverable from known *local* logic features such as canalization and effective connectivity, so that the transfer result merely exploits the fact that real GRNs are heavily canalizing. Two experiments argue against this reading (Table V, Fig. 2c).

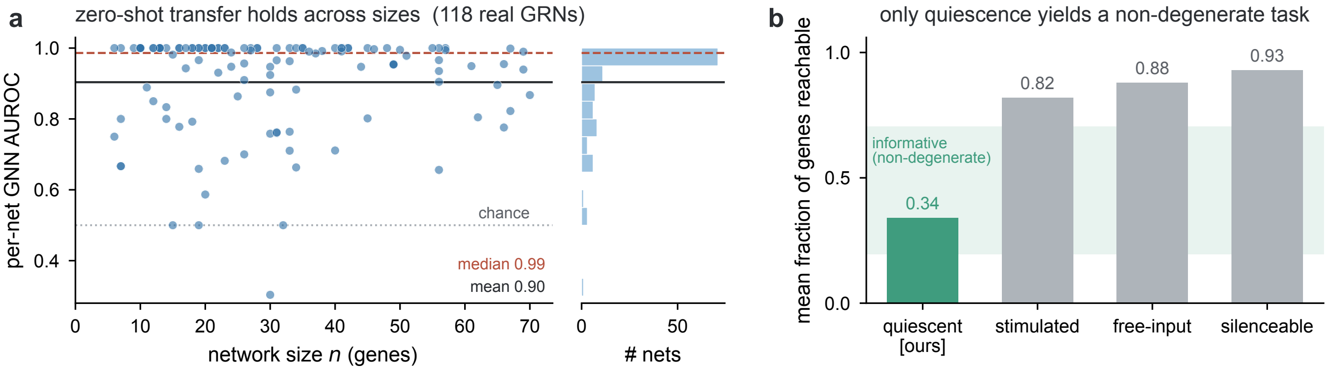

Fig. 3. Zero-shot transfer to real GRNs, and why the quiescent task is principled. **a**, Per-net AUROC of the synthetic-trained GNN on the 118 real BBM networks vs network size (left), with the left-skewed distribution (right; median 0.99, mean 0.90); transfer shows no significant monotone size trend, though the failure tail falls in the size-extrapolation regime. **b**, Mean fraction of genes reachable under four driving conditions: only the quiescent inputs-OFF task we study (green) avoids saturation and falls in the informative band, so it is the principled non-degenerate choice rather than a cherry-picked one.

First, an ablation of three GNN variants: *sign* (separate $\pm$ transforms), *no-sign* (message passing, sign ignored), and *MLP* (no message passing, a neural model on the per-node features, i.e. a neural canalization baseline). In distribution, message passing adds $+0.026$ over the MLP and sign-awareness a further $+0.021$; on real GRNs the effect sharpens: sign-awareness adds $+0.091$, with the sign-agnostic GNN collapsing to the feature-baseline level. Propagation, and the sign of regulation, are what the GNN exploits.

Second, a dedicated CANA effective-connectivity tree [12]: per-input activity $a_i = \Pr_x[f(x) \neq f(x \oplus e_i)]$, effective connectivity $k_e = \sum_i a_i$, input redundancy $r = 1 - k_e/k$, the activity distribution, canalizing depth, and bias. The GNN beats this strengthened baseline by $+0.097$ in distribution; on transfer the CANA tree drops to AUROC 0.465, slightly below chance. We read the below-chance value not as evidence that canalization is irrelevant to the dynamics (it is not), but as a *distribution-shift artifact*: the local canalization→activatability relation learned on the synthetic ensemble inverts on real GRNs, so a model relying only on those local features fails to port. The substantive comparison is therefore the in-distribution gap ($+0.097$, where no shift confounds it) together with the sign / no-sign ablation, which jointly indicate that the GNN's *transferable* signal is sign-aware propagation rather than a re-encoding of local canalization.

TABLE V
PER-NODE ACTIVATION REACHABILITY: THE GNN VS CANALIZATION BASELINES (MEAN±STD OVER 3 TRAINING SEEDS). THIS ABLATION HARNESS USES A LARGER SYNTHETIC-TRAINING BUDGET THAN THE MAIN EXPERIMENT OF §VII-B, SO ITS ABSOLUTE AUROC VALUES RUN SLIGHTLY HIGHER (BOTH IN DISTRIBUTION AND ON TRANSFER) WHILE THE RANKINGS AGREE; THE SIGN-AWARE TRANSFER FIGURE HERE (0.911) IS CONSISTENT WITH THE CLUSTER-BOOTSTRAP HEADLINE (0.906, 95% CI [0.879, 0.932]). THE MLP VARIANT IS AN IN-DISTRIBUTION DIAGNOSTIC AND WAS NOT EVALUATED ON TRANSFER (—).

method	in-distribution	transfer (real)
GNN (sign-aware)	$\mathbf{0.984 \pm 0.001}$	$\mathbf{0.911 \pm 0.010}$
GNN (sign-agnostic)	0.963 ± 0.002	0.820 ± 0.022
GNN (MLP, no msg. passing)	0.937 ± 0.002	—
feature tree	0.929 ± 0.003	0.814 ± 0.003
CANA effective-connectivity tree	0.887 ± 0.002	0.465 ± 0.009

TABLE VI
TRIVIAL-BIAS CONTROL (REACHABILITY AUROC).

predictor	all-ones target	non-trivial target
mean-output-bias only	0.716	0.671
full logic features	0.921	0.897

D. Controls

Beyond the chaos control (Table III) and the shuffle-null and leave-graph-out checks, we verify the reachability signal is not a trivial output-bias artifact (Table VI). Here the prediction target is whether a designated state is reachable from quiescence: the *all-ones target* is the all-ON state (tracking the per-node activation task), and the *non-trivial target* is a fixed non-constant state (the alternating 0101... bit pattern), whose base rate is less aligned with mean output bias. A mean-output-bias-only predictor reaches just 0.716 versus 0.921 for the full model on the all-ones target, and predictability persists (0.897) on the non-trivial target where the bias predictor is even weaker (0.671). All transfer splits are de-confounded for the random/canalizing logic mix.

E. Application: the budding-yeast cell cycle

To make the predictions concrete, we apply the synthetic-trained GNN *zero-shot* to the budding-yeast cell-cycle network, a classic Boolean GRN model in the BBM collection [19] ($n = 20$), and ask which genes are activatable from G1 quiescence (the all-OFF state). The predictor attains AUROC 0.91 over all 20 genes (thresholded accuracy 0.70, 14/20 genes correct): it ranks the core cyclins and regulators (Clb2, Clb5, Cdh1, Sic1, Whi5) as activatable and the spindle and budding checkpoints (SpindleCP, BuddingCP), the Start transition, the Cdc14 phosphatase, and the G1 cyclin Cln2 as *not* spontaneously activatable from quiescence (Table VII), which is qualitatively consistent with cell-cycle entry requiring

TABLE VII
ZERO-SHOT PER-GENE ACTIVATION-REACHABILITY PREDICTIONS ON THE BUDDING-YEAST CELL CYCLE (SELECTED GENES; AUROC 0.91 OVER ALL 20).

gene	GNN p(activatable)	exact label
Clb2	1.00	activatable
Clb5	1.00	activatable
Cdh1	1.00	activatable
Sic1	1.00	activatable
Whi5	1.00	activatable
Cdc14	0.18	not
SpindleCP	0.01	not
BuddingCP	0.01	not
Start	0.01	not
Cln2	0.00	not

size and signal cues rather than arising spontaneously. The errors are intermediate regulators near the activatable boundary (e.g. Cdc20, Cln3, FEAR). This is an *illustration* of the prediction, not an independent validation: the labels here are themselves from the exact oracle, so the vignette shows what the zero-shot predictor recovers on a familiar network rather than establishing a new biological finding. As a single 20-gene network, the reported AUROC is an illustrative point estimate without a confidence interval.

VIII. Discussion and Limitations

The GNN's advantage is *task-specific*: it wins on reachability, the propagation-shaped property, and not on the global has-cyclic property (§VII-B); we do not claim a universal dynamics predictor, and view the match between message-passing locality and reachability propagation as the explanation. Per-node reachability difficulty depends strongly on the *driving condition*, and the quiescent activation task we study is the principled non-degenerate choice rather than a hand-picked one. We verified this on the real BBM networks: the all-OFF / inputs-OFF convention is well-behaved (pooled base 0.34), whereas the natural alternatives all *saturate* (Fig. 3b): a free-input "activatable under some signal" variant (mean base 0.88), a fully-stimulated condition (inputs fixed ON; mean per-net base 0.82), and a silenceability condition (reaching 0 from the all-ON state; mean base 0.93) each make almost every gene reachable, leaving little to predict. The quiescent condition is thus not cherry-picked for learnability but is the one biologically meaningful condition that yields a non-degenerate label; characterizing the framework under richer, non-saturating multi-signal conditions is important future work.

There is a genuine synthetic→real distribution shift (synthetic activation base rate 0.86 vs real 0.34); the GNN transfers in spite of it, which is its main strength, although a synthetic ensemble matched to GRN statistics (scale-free wiring, realistic nested-canalizing prevalence) could shrink the shift and is promising future work. Finally, exact synthetic ground truth limits training networks to $n \leq 16$. We stress that on the curated networks studied here the symbolic oracle is fast (it labels almost all of them in well under a second), so the contribution is a scientific one, namely that the logic carries transferable reachability structure, rather than a computational speed-up; we make no efficiency claim over the exact solver in this regime.

IX. Conclusion

The wiring diagram is necessary but far from sufficient for a Boolean network's dynamics. We have used that classical fact as the starting point for a well-posed learning problem with free, exact supervision. A sign-aware, size-agnostic GNN that reads the local logic predicts asynchronous reachability, and it beats an all-truth-table-bits model exactly where propagation matters. Trained only on small synthetic networks, the same model transfers zero-shot to real gene-regulatory networks and outperforms every size-agnostic baseline, including a canalization model that does not transfer at all. The logic, then, carries learnable reachability structure that a graph-only model cannot recover (established in distribution, §V) and that transfers across networks, and a propagation-matched architecture is what brings it within reach.

Reproducibility

All labels are computed by exact symbolic tools (state-transition-graph enumeration; `biodivine_aeon` BDD reachability); no human annotation or external API is used. The synthetic generators, the GNN, and all baselines run on a single GPU; the curated networks are the public biodivine-boolean-models collection. All code, trained models, and the symbolic label cache (sufficient to reproduce every reported number, including the bootstrap CIs) are available at https://github.com/Beiciccc/attractornet.

References

[1] S. A. Kauffman, "Metabolic stability and epigenesis in randomly constructed genetic nets," *J. Theor. Biol.*, vol. 22, no. 3, pp. 437–467, 1969.

[2] R. Thomas, "Boolean formalization of genetic control circuits," *J. Theor. Biol.*, vol. 42, no. 3, pp. 563–585, 1973.

[3] A. Richard and J.-P. Comet, "Necessary conditions for multistationarity in discrete dynamical systems," *Discrete Appl. Math.*, vol. 155, no. 18, pp. 2403–2413, 2007.

[4] A. Richard, "Positive and negative cycles in Boolean networks," *J. Theor. Biol.*, vol. 463, pp. 67–76, 2019.

[5] J. Aracena, "Maximum number of fixed points in regulatory Boolean networks," *Bull. Math. Biol.*, vol. 70, no. 5, pp. 1398–1409, 2008.

[6] F. Bridoux, A. Durbec, K. Perrot, and A. Richard, "Complexity of fixed point counting problems in Boolean networks," *J. Comput. Syst. Sci.*, vol. 126, pp. 138–164, 2022.

[7] P. Sil, A. Subbaroyan, S. Kulkarni, O. C. Martin, and A. Samal, "Biologically meaningful regulatory logic enhances the convergence rate in Boolean networks and bushiness of their state transition graph," *Brief. Bioinform.*, vol. 25, no. 3, art. bbae150, 2024.

[8] D. Rumiantsau, A. Lesne, and M.-T. Hütt, "Predicting attractors from spectral properties of stylized gene regulatory networks," *Phys. Rev. E*, vol. 108, no. 1, art. 014402, 2023.

[9] F. M. Weidner, J. D. Schwab, S. D. Werle, N. Ikonomi, L. Lausser, and H. A. Kestler, "Capturing dynamic relevance in Boolean networks using graph-theoretical measures," *Bioinformatics*, vol. 37, no. 20, pp. 3530–3537, 2021.

[10] A. Mizera and J. Zarzycki, "The GATTACA framework: Graph neural network-based reinforcement learning for controlling biological networks," in *Proc. AAAI Conf. Artif. Intell. (AAAI-26)*, vol. 40, no. 2, 2026, pp. 873–880.

[11] B. Kang, H. Kumar, M. Lee, B. Chakraborty, and S. Mukhopadhyay, "Learning locally interacting discrete dynamical systems: Towards data-efficient and scalable prediction," in *Proc. 6th Annu. Learn. Dyn. Control Conf. (L4DC)*, Proc. Mach. Learn. Res., vol. 242, 2024, pp. 1357–1369.

[12] M. Marques-Pita and L. M. Rocha, "Canalization and control in automata networks: Body segmentation in *Drosophila melanogaster*," *PLoS ONE*, vol. 8, no. 3, art. e55946, 2013.

[13] N. Beneš, L. Brim, O. Huvar, S. Pastva, D. Šafránek, and E. Šmijáková, "AEON.py: Python library for attractor analysis in asynchronous Boolean networks," *Bioinformatics*, vol. 38, no. 21, pp. 4978–4980, 2022.

[14] N. Beneš, L. Brim, J. Kadlecaj, S. Pastva, and D. Šafránek, "AEON: Attractor bifurcation analysis of parametrised Boolean networks," in *Computer Aided Verification (CAV 2020)*, LNCS, vol. 12224. Springer, 2020, pp. 569–581.

[15] V.-G. Trinh, K. H. Park, S. Pastva, and J. C. Rozum, "Mapping the attractor landscape of Boolean networks with biobalm," *Bioinformatics*, vol. 41, no. 5, art. btaf280, 2025.

[16] B. Derrida and Y. Pomeau, "Random networks of automata: a simple annealed approximation," *Europhys. Lett.*, vol. 1, no. 2, pp. 45–49, 1986.

[17] M. Schlichtkrull, T. N. Kipf, P. Bloem, R. van den Berg, I. Titov, and M. Welling, "Modeling relational data with graph convolutional networks," in *The Semantic Web (ESWC 2018)*, LNCS, vol. 10843. Springer, 2018, pp. 593–607.

[18] T. Derr, Y. Ma, and J. Tang, "Signed graph convolutional networks," in *Proc. IEEE Int. Conf. Data Mining (ICDM)*, 2018, pp. 929–934.

[19] R. G. Todd and T. Helikar, "Ergodic sets as cell phenotype of budding yeast cell cycle," *PLoS ONE*, vol. 7, no. 10, art. e45780, 2012.